Methodology

Self-contained methodology for the BIOAI-2026 submission. Describes the implemented pipeline (what we did and how), the feasibility experiments run so far, and the planned extensions (what we will do), including the learned end-to-end error detector and the move to a larger training corpus. Completed work is written in the past tense; planned work in the future tense.

1. Problem formulation

We treat brain-tumour segmentation reliability as a **case-level prediction task**: given an automatic segmentation and the model's own stochastic behaviour — but **no reference mask** — decide whether the output's whole-tumour volume error is large enough to be clinically consequential, so that unreliable cases are routed to expert review while the rest are safely automated. The study is deliberately a *triage benchmark*: we do not propose a new segmentation architecture or a new uncertainty estimator, but ask which readily available, reference-free signals actually predict consequential volume error, under a protocol fixed in advance.

2. Data and patient-level splits

All experiments use publicly available, expert-annotated brain-tumour MRI. For the feasibility stage reported here we used the **MSD Task01_BrainTumour** collection (484 patients, four co-registered modalities each). The identical pipeline targets larger glioma corpora (§9) through a configuration switch that carries each dataset's label convention and file layout.

Each patient was assigned to exactly one of four disjoint splits (`scripts/00_make_splits.py`), so no patient appears in more than one:

Split	Share	Purpose
`seg_train`	45 %	train the segmenter only
`seg_val`	5 %	select the segmenter checkpoint; report its Dice
`risk_dev`	30 %	develop risk models (repeated cross-validation), fix thresholds
`risk_test`	20 %	evaluate once, after the protocol is committed

This separation is essential to validity: the segmenter never sees the risk cases, so the errors the risk model learns from are realistic rather than the unrealistically small errors a model makes on its own training data. Datasets that share source scans (e.g. MSD and BraTS) are never combined.

3. Segmentation model

The segmenter is a standard residual 3D U-Net (MONAI) with dropout in every unit, trained once with fixed hyper-parameters (96^3 patches, batch size 1, mixed-precision, Dice loss, Adam, learning rate 2×10^{-4}) and no architecture search. We report its Dice against published BraTS numbers so that reviewers can confirm it is a reasonable, not deliberately weakened, baseline. Dropout in every unit is what makes Monte-Carlo dropout a usable uncertainty source on a single 6 GB GPU, avoiding the cost of a deep ensemble.

4. Reference-free stochastic inference

For each case we drew multiple stochastic predictions with MC-dropout kept active, each pass additionally combined with a random axis flip (test-time augmentation), so that the passes cost the same as plain MC-dropout. From the per-voxel **mean** and **variance** across passes we computed all downstream signals, for the three nested BraTS regions — whole tumour (WT), tumour core (TC), enhancing tumour (ET). No reference mask enters this stage.

5. Case-level, reference-free features

From the stochastic probability maps we derived per-region features requiring no ground truth, organised into the families that the comparison contrasts:

- `uncertainty` — mean entropy and variance inside the predicted mask and a dilated ROI, and total entropy volume (standard voxel-uncertainty summaries aggregated to the case level);
- `disagreement` — coefficient of variation and relative range of the per-pass predicted volume, and the mean/minimum Dice between each pass and the consensus mask (how much the volume estimate moves across passes);
- `boundary_morphology` — boundary-band (shell) entropy and variance; the *undecided volume fraction* (voxels with probability between 0.2 and 0.8, relative to predicted volume) and the soft-minus-hard volume difference, both direct proxies for how far the volume estimate could move; and shape descriptors (connected components, largest-component fraction, number of small components, surface-to-volume ratio) with log predicted volume;
- `combined` — all of the above across all regions plus nested-region consistency terms (ET outside TC, TC outside WT).

Two reference points anchor every comparison: `volume_only` (log predicted volume alone — small tumours have large relative errors) and `best_single` (the single strongest uncertainty score, selected on `risk_dev`), i.e. the "just threshold one uncertainty number" baseline.

6. Pre-specified error endpoints

The analysis protocol was fixed and committed before any evaluation on the test set (`configs/preregistration.yaml`; the commit timestamp is the evidence, and the evaluation script refuses to run with uncommitted changes to it). The **primary endpoint** labels a case consequential when the whole-tumour relative volume error is ≥ 25 % **and** the absolute error is ≥ 1 mL: the 25 % links to the RANO progression threshold, and the 1 mL floor prevents a sub-millilitre miss on a tiny lesion from being counted. Reference volume error is measured from the expert mask. Secondary endpoints (WT at 10 % and 40 %, TC at 25 %, and WT Dice < 0.80) are reported in full whatever they show and are not used to pick the primary result.

7. Risk models and comparison

The primary model is L2-regularised logistic regression (class-balanced, median imputation, standardised features); shallow histogram gradient boosting is the secondary model. On `risk_dev` we obtained risk scores by repeated (5×) stratified 5-fold cross-validation (out-of-fold scores), which we used to compare feature families, choose `best_single`, and fix

the operating threshold at 90 % sensitivity. The frozen models and threshold were then applied **once** to `risk_test`. The pre-specified success criterion was that a candidate family (`boundary_morphology` or `combined`) beat **both** baselines on a paired bootstrap of AUROC; the outcome was reported whether or not it held.

8. Evaluation

We report AUROC (primary metric) and AUPRC shown against prevalence, each with 95 % bootstrap confidence intervals (2,000 resamples). As a triage mechanism we report, at the fixed-sensitivity operating point, the sensitivity actually achieved, the false-alert rate, the fraction of cases retained for automation, and the residual consequential-error rate among retained cases, together with the risk–coverage curve and its area and the Brier score (read with the caveat that a class-balanced logistic model is not calibrated). Feature families are compared with a paired bootstrap of AUROC, reporting the difference, its 95 % CI, and a two-sided p-value.

9. Planned extensions (future work)

The feasibility stage established a working, pre-registered pipeline and an honest baseline comparison. On that feasibility dataset the engineered features did not outperform the simple baselines on the primary endpoint, which motivates the following planned work; none of it alters the frozen endpoint.

1. **Scale to a large training corpus.** End-to-end learning and reliable effect estimates require many more cases than MSD provides. We will run the identical protocol on a large, expert-annotated glioma corpus (candidate: BraTS 2023 Adult Glioma, 1,251 cases, extendable toward the ~2,200-case BraTS 2024 cohort, and optionally pooled with other whole-tumour-annotated collections), giving roughly an order of magnitude more development cases and a lower, more realistic prevalence of consequential error.

1. **A learned, end-to-end error detector.** In place of hand-crafted features we will train a small, heavily-regularised 3D convolutional network that reads the segmenter's uncertainty volumes directly (per-region mean probability, variance, and entropy, resampled to a fixed low-resolution grid) and outputs a single case-level risk score. The network is deliberately tiny (on the order of 10^4 parameters) with dropout, weight decay, and flip augmentation, because positives are scarce. Crucially, it is evaluated under the *same* pre-registered protocol — the same repeated cross-validation on `risk_dev`, the same fixed-sensitivity threshold, and the same paired-bootstrap comparison against `volume_only` and `best_single` — so that "learned" enters as one more family on equal terms. The learned detector is judged only on a fresh, once-opened test split; the existing feasibility test set is not reused.

1. **Cross-fitting for more positives.** Training several segmenters on disjoint folds so that every case receives an out-of-fold prediction roughly doubles the number of error positives available to the risk model, tightening confidence intervals without lowering the clinical threshold.

1. **Direction and magnitude of error.** Beyond the binary flag, we will test whether the same reference-free signals indicate whether a case is over- or under-segmented and by how much, moving from error *detection* toward reference-free volume *correction*.

1 **External validation and calibration.** We will validate on an out-of-distribution cohort (e.g. BraTS-Africa: lower field strength, different population) and add probability calibration so that outputs can be read as risks.

1 **Annotation-noise context.** We will cite measured inter-rater *volume* variability for BraTS and state where the 25 % threshold sits relative to it, since our feasibility results suggest the engineered signals help mainly once the error exceeds the annotation-noise floor (large errors and overlap failures).